

Semester DB Project: Phase III

CSC 436 | Research Group Database Application

Part I: Review Feedback & Revision Plan

Strengths Highlighted in Feedback

Feedback from the Database Expo was broadly positive. Reviewers consistently praised the role-based access control system, noting that limiting functionality based on user account type was a meaningful and well-executed security feature. The Win98/Internet Explorer aesthetic was called out as a standout design choice, with multiple reviewers describing it as creative and well-executed.

Reviewers also appreciated the overall organization of the application, describing it as focused, clean, and purposeful. The security section of the presentation was specifically commended as detailed and thorough, and the flowchart depicting the login and edit process was noted as a clear and useful visual.

Weaknesses and Suggested Improvements

Three distinct improvement themes emerged from reviewer feedback:

- **Search and query functionality:** Three separate reviewers noted that the application required users to go through the command line or SQL to perform any kind of search or data lookup. One reviewer wrote that adding query functionality to the website would help users who do not have programming or CLI experience. Another simply wrote to add a search bar. A third echoed this, stating that adding a way to search without the command line would help lots of users since not everybody knows SQL.
- **Deployment:** Two reviewers mentioned that the application was running locally rather than being hosted live, and suggested moving to cPanel or a live web host. This was addressed separately during development and the application is now fully deployed at keelyfowler.rhody.dev/researchdb.
- **View logic:** One reviewer from Phase II flagged that the `project_access_view` may not be behaving as intended, suggesting the relationship logic should be reviewed to ensure it accurately reflects access permissions.

Planned Revisions

Based on the feedback, the group identified four meaningful revisions to implement before final submission. Three were planned based on reviewer feedback; a fourth was discovered during a code audit conducted while implementing the other changes. Each revision is tied directly to reviewer feedback or a confirmed technical issue, and none were carried forward from prior phases.

- Add a global search feature to the application, addressing the most repeated piece of feedback across all reviewers.
- Add performance indexes on frequently filtered columns in the database, converting a previously listed future improvement into an actual implementation.
- Fix a confirmed logic bug in the project_access_view that caused it to return incorrect results regardless of a user's project role.
- Fix a missing FROM keyword in member_view.php discovered during a code audit, which caused a fatal SQL syntax error on all member profile pages.

Part II: Implementation of Revisions

Revision 1: Global Search Feature (search.php)

What Was Changed

A new page, search.php, was added to the application. A search input was also added to the quick-links navigation bar on every page, allowing users to search from anywhere without navigating away first. The search covers five entity types: Projects, Tasks, Members, Experiments, and Literature. Each entity searches across multiple relevant fields — literature searches title, journal, and year; members searches name, email, and role via a join on person_group; experiments searches title, objective, and description. Status filter dropdowns were also added to projects.php and tasks.php, allowing users to filter records by status without writing any SQL.

Why the Change Was Made

This directly addresses the most repeated piece of feedback from the Database Expo. Three reviewers independently noted that performing any kind of lookup required going through the command line or running raw SQL, which is not a realistic expectation for the intended users of a research group management tool. Researchers are not database administrators, and requiring CLI access to find a project or paper creates a significant usability barrier.

How It Improves the Application

The search page runs five separate prepared statements simultaneously across the project, task, Person, experiment, and literature tables. Results are returned in grouped, collapsible sections, one per entity type, each with a result count in the header. Sections with results expand by default; sections with zero results collapse automatically. Each result row links to the corresponding detail view page. The search term appears in the browser address bar via GET, making results linkable and bookmarkable. All queries use parameterized LIKE bindings with unique parameter names, so no raw user input ever touches SQL.

← → ↻ keelyfowler.rhody.dev/researchdb/Webapp/projects.php?status=Completed ☆ 🔍 ⬇️ K School ⋮

110 TA MTH 215 CSC310 PHY203 CSF 202 211 TA TA Setup- - - Google... ☐

⌵ All Bookmarks

Projects – Research Group Database – Microsoft Internet Explorer

File Edit View Favorites Tools Help

⏪ ⏩ ↺ 🏠 🗑️

Back Forward Refresh Home Log Out

Address http://localhost/researchdb/projects.php?status=Completed Go

Projects

Logged in as: Alice Johnson
Role: Principal Investigator

✓ Experiments Projects Members Literature ✓ Tasks Departments

Status

All
Active
✓ Completed
Cancelled
On Hold

 Search...

+ New Project

#	Title	Status	Start Date	End Date	Actions
29	Flu Mobility Surveillance	Completed	2023-03-01	2024-02-29	Edit Delete
25	Rare Earth Isotope Profiling	Completed	2022-09-01	2024-08-31	Edit Delete
24	Migratory Bird Tracking Study	Completed	2021-06-01	2023-05-31	Edit Delete
18	Human Migration Genomics	Completed	2022-03-01	2024-02-28	Edit Delete
16	Deep Sea Carbon Flux Study	Completed	2021-05-01	2023-04-30	Edit Delete
12	COVID-19 Variant Surveillance	Completed	2022-01-01	2023-12-31	Edit Delete
10	Drug-Receptor Binding Simulation	Completed	2022-04-01	2024-03-31	Edit Delete
7	Social Media Sentiment Mining	Completed	2023-05-15	2024-05-14	Edit Delete
4	Arctic Ice Melt Simulation	Completed	2021-09-01	2023-08-31	Edit Delete
3	AI Tumor Detection	Completed	2023-03-01	2024-09-30	Edit Delete

✓ Done research_group_db @ localhost Local intranet

← → ↻ keelyfowler.rhody.dev/researchdb/Webapp/search.php?q=CRISPR ☆ 🔍 ⬇️ K School ⋮

110 TA MTH 215 CSC310 PHY203 CSF 202 211 TA TA Setup- - - Google... ☐

⌵ All Bookmarks

Search – Research Group Database – Microsoft Internet Explorer

File Edit View Favorites Tools Help

⏪ ⏩ ↺ 🏠 🗑️

Back Forward Refresh Home Log Out

Address http://localhost/researchdb/search.php?q=CRISPR Go

Search

Logged in as: Alice Johnson
Role: Principal Investigator

✓ Experiments Projects Members Literature ✓ Tasks Departments

Search CRISPR

CRISPR Search

Projects (1 result)

Title	Status	Start Date
CRISPR Gene Editing Study	Active	2023-01-15

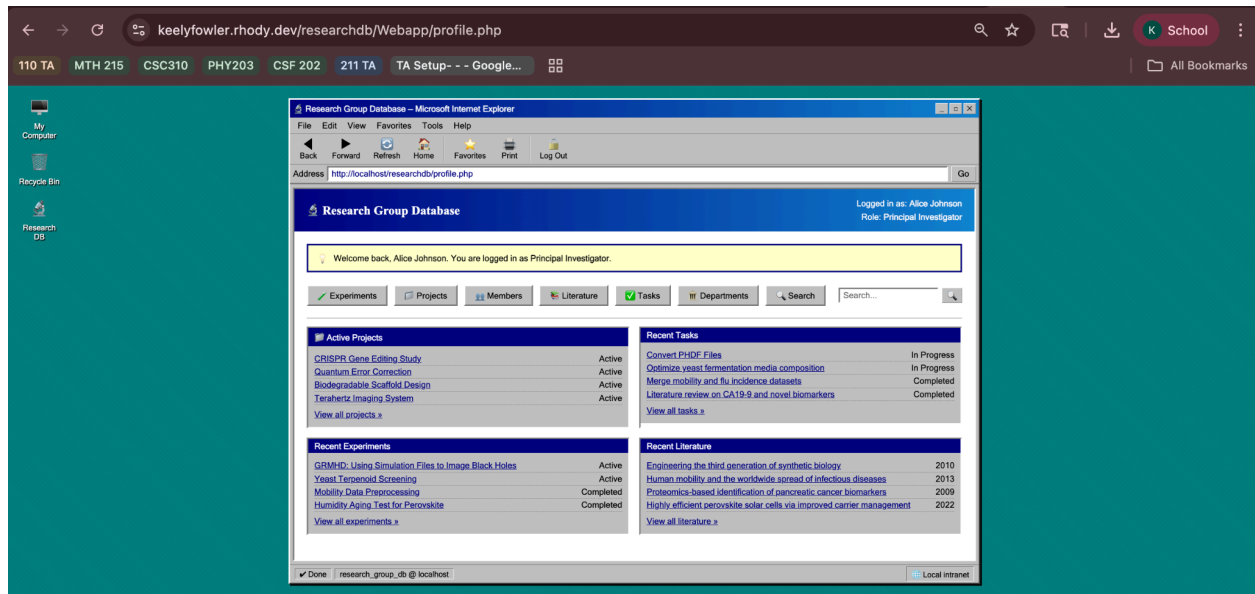
Tasks (0 results)

Experiments (1 result)

Title	Status	Start Date
CRISPR Transfection Trial 1	Completed	2023-02-01

Literature (0 results)

✓ Done research_group_db @ localhost Local intranet



SQL Queries Used

-- Projects

```
SELECT project_id, title, description, status, start_date
FROM project
WHERE title LIKE :term OR description LIKE :term
```

-- Tasks

```
SELECT task_id, title, description, status, progress
FROM task
WHERE title LIKE :term OR description LIKE :term
```

-- Members

```
SELECT DISTINCT p.ID, p.name, p.email, p.role AS sys_role, pg.role AS group_role
FROM Person p
LEFT JOIN person_group pg ON p.ID = pg.person_id
WHERE p.name LIKE :t1 OR p.email LIKE :t2 OR pg.role LIKE :t3
```

Note: The members query uses distinct parameter names (:t1, :t2, :t3) because PDO does not allow the same named parameter to appear more than once in a single prepared statement.

-- Experiments

```
SELECT experiment_id, title, objective, status, start_date
FROM experiment
WHERE title LIKE :term OR objective LIKE :term
```

-- Literature

```
SELECT lit_id, title, journal, year
```

FROM literature
WHERE title LIKE :term OR journal LIKE :term

Revision 2: Performance Indexes on Frequently Filtered Columns

What Was Changed

Six indexes were added to research_group_db_final.sql on columns that are frequently used in WHERE clauses and ORDER BY expressions across the application but had no index beyond the default primary key.

```
CREATE INDEX idx_project_status ON project(status);  
CREATE INDEX idx_task_status ON task(status);  
CREATE INDEX idx_task_due_date ON task(due_date);  
CREATE INDEX idx_task_progress ON task(progress);  
CREATE INDEX idx_literature_year ON literature(year);  
CREATE INDEX idx_literature_journal ON literature(journal);
```

The screenshot displays the phpMyAdmin interface. On the left, a sidebar shows the database structure with 'keelyfow' as the selected database and 'Tables' as the selected category. The main panel shows the 'Table structure' view for the 'project' table. The table has six columns: 'project_id' (int(11), PRIMARY), 'title' (varchar(100)), 'description' (text), 'start_date' (date), 'end_date' (date), and 'status' (varchar(50)). Below the table structure, there is a section for 'Indexes' showing two indexes: 'PRIMARY' (BTREE, Unique, Packed, Column: project_id, Cardinality: 30) and 'idx_project_status' (BTREE, Unique, Packed, Column: status, Cardinality: 3). At the bottom, there is a form to 'Create an index on' 1 column(s).

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	project_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	title	varchar(100)	utf8_unicode_ci		No	None			Change Drop More
3	description	text	utf8_unicode_ci		Yes	NULL			Change Drop More
4	start_date	date			Yes	NULL			Change Drop More
5	end_date	date			Yes	NULL			Change Drop More
6	status	varchar(50)	utf8_unicode_ci		No	None			Change Drop More

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	project_id	30	A	No	
Edit Rename Drop	idx_project_status	BTREE	No	No	status	3	A	No	

phpMyAdmin

Recent Favorites

keelyfow

keelyfow_research_group_db

Tables

New

department

department_test

experiment

expt_literature

group_project

literature

Person

person_dept

person_group

project

project_expt

project_literature

project_member

project_tasks

research_dept

research_group

task

task_assignment

Views

keelyfow_toystore

Server: localhost:3306 » Database: keelyfow_research_group_db » Table: task

Browse Structure SQL Search Insert Export Import Operations Triggers

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	task_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	title	varchar(100)	utf8_unicode_ci		No	None			Change Drop More
3	description	text	utf8_unicode_ci		Yes	NULL			Change Drop More
4	status	varchar(50)	utf8_unicode_ci		No	None			Change Drop More
5	due_date	date			Yes	NULL			Change Drop More
6	completed_date	date			Yes	NULL			Change Drop More
7	progress	int(11)			No	0			Change Drop More

Check all With selected: Browse Change Drop Primary Unique Index Spatial

Print Propose table structure Move columns Normalize

Add 1 column(s) after progress Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	task_id	31	A	No	
Edit Rename Drop	idx_task_status	BTREE	No	No	status	2	A	No	
Edit Rename Drop	idx_task_due_date	BTREE	No	No	due_date	26	A	Yes	
Edit Rename Drop	idx_task_progress	BTREE	No	No	progress	11	A	No	

Create an index on 1 columns Go

phpMyAdmin

Recent Favorites

keelyfow

keelyfow_research_group_db

Tables

New

department

department_test

experiment

expt_literature

group_project

literature

Person

person_dept

person_group

project

project_expt

project_literature

project_member

project_tasks

research_dept

research_group

task

task_assignment

Views

keelyfow_toystore

Server: localhost:3306 » Database: keelyfow_research_group_db » Table: literature

Browse Structure SQL Search Insert Export Import Operations Triggers

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	lit_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	title	varchar(255)	utf8_unicode_ci		No	None			Change Drop More
3	year	year(4)			No	None			Change Drop More
4	journal	varchar(255)	utf8_unicode_ci		Yes	NULL			Change Drop More
5	doi	varchar(255)	utf8_unicode_ci		Yes	NULL			Change Drop More
6	url	varchar(255)	utf8_unicode_ci		Yes	NULL			Change Drop More
7	theory	text	utf8_unicode_ci		Yes	NULL			Change Drop More

Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext

Print Propose table structure Move columns Normalize

Add 1 column(s) after theory Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	lit_id	30	A	No	
Edit Rename Drop	idx_literature_year	BTREE	No	No	year	21	A	No	
Edit Rename Drop	idx_literature_journal	BTREE	No	No	journal	20	A	Yes	

Create an index on 1 columns Go

Why the Change Was Made

The final report from Phase II explicitly listed adding indexes on frequently used columns as a future improvement. The columns selected are those that appear most often in the application's WHERE clauses: status is filtered on projects and tasks throughout the interface, due_date and progress drive the task prioritization query, and year and journal are the most natural lookup fields on the literature table. With the addition of the search feature in Revision 1, these columns are now

queried even more frequently via LIKE, making the indexes more impactful than when they were originally listed as a future item.

How It Improves the Application

Without indexes, MySQL performs a full table scan on every query that filters by these columns. The indexes allow the database engine to locate matching rows directly, reducing query time as the dataset grows. While the current dataset is small enough that the difference is not perceptible, this revision demonstrates correct database design practice and converts a previously acknowledged gap into a concrete implementation.

Additional Finding During Testing

During reimport testing, the member_contact_view and project_access_view definitions were found to be ordered before the Person table definition in the SQL file, causing a fatal import error on fresh import. Both view definitions were relocated to after the last CREATE TABLE statement to resolve this.

Revision 3: Fix Logic Bug in project_access_view

What Was Changed

The JOIN condition in project_access_view was rewritten. The original view used a single JOIN with an OR condition that made one of its branches always evaluate to true, causing the view to return the same results for all roles regardless of whether the user was a Project Lead or a standard member.

Original (buggy) definition:

```
JOIN project proj ON (  
    (pm.role = 'Project Lead')  
    OR (proj.project_id = pm.project_id)  
)
```

The second condition, proj.project_id = pm.project_id, is always true for any row returned by the preceding join on project_member. This means every user matched every project they were connected to regardless of their role — the Project Lead branch never had any effect.

The corrected view uses a UNION to separate the two access paths cleanly:

```
CREATE VIEW project_access_view AS  
-- All members see projects they are directly assigned to  
SELECT p.ID AS person_id, p.name, pm.role AS project_role,  
    proj.project_id, proj.title, proj.description,  
    proj.status, proj.start_date, proj.end_date
```



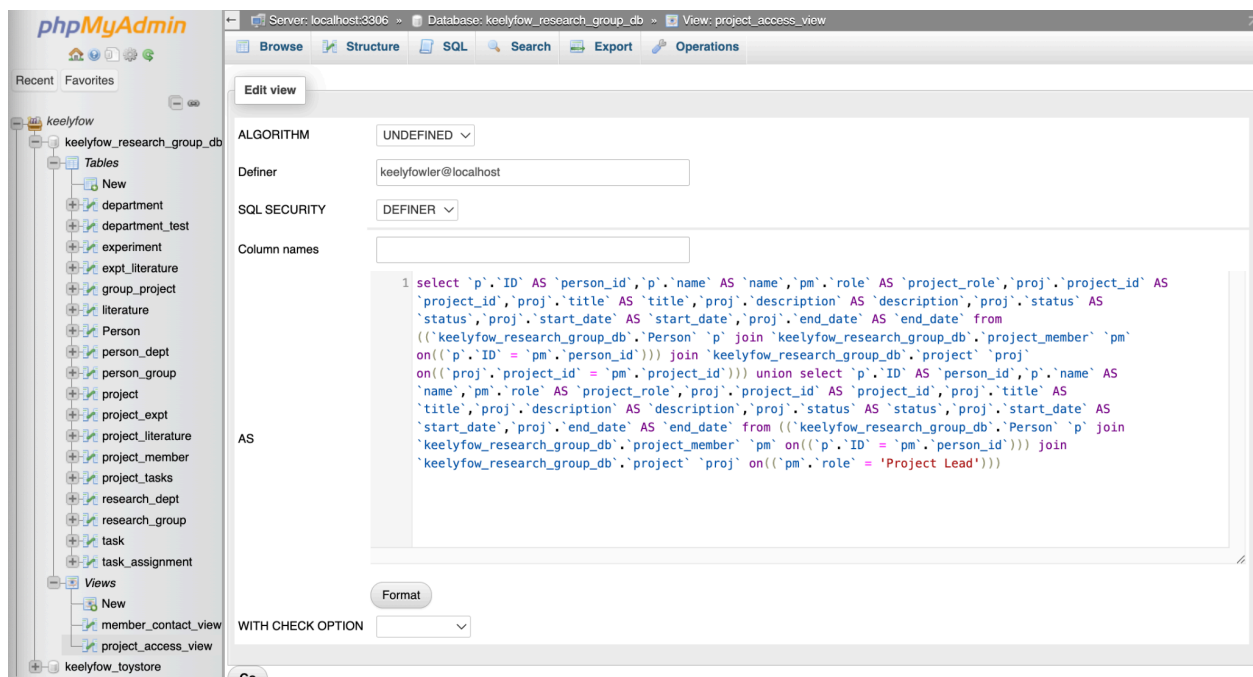
```

FROM person p
JOIN project_member pm ON p.ID = pm.person_id
JOIN project proj ON proj.project_id = pm.project_id

UNION

-- Project Leads additionally see ALL projects
SELECT p.ID AS person_id, p.name, pm.role AS project_role,
      proj.project_id, proj.title, proj.description,
      proj.status, proj.start_date, proj.end_date
FROM person p
JOIN project_member pm ON p.ID = pm.person_id
JOIN project proj ON pm.role = 'Project Lead';

```



Why the Change Was Made

A Phase II reviewer specifically noted that view logic should be reviewed to ensure all relationships behave as intended. When the view was examined against the SQL, the OR condition was confirmed to make the Project Lead branch unreachable in practice. A standard member and a Project Lead would receive identical result sets, which contradicts the intended access control design described in the final report.

How It Improves the Application

The corrected view now accurately enforces the intended access model. Standard members see only the projects they are assigned to. Project Leads see all projects in the system, consistent with their

elevated role. The fix brings the view into alignment with the database role system and access control model built in Phase II.

Revision 4: Fix Missing FROM Keyword in member_view.php

What Was Changed

During a code audit performed while investigating the view ordering issue, member_view.php line 9 was found to contain `SELECT * Person WHERE ID = :id` — missing the FROM keyword, which causes a fatal SQL syntax error whenever any user attempts to view a member profile page. The query was corrected to `SELECT * FROM Person WHERE ID = :id`.

Why the Change Was Made

Discovered during a code audit performed while verifying table and column name consistency across the application. While the error had not consistently surfaced during prior testing, a missing FROM keyword is a hard SQL syntax violation that fails on any standard MySQL execution.

How It Improves the Application

Member profile pages now load correctly and reliably. Clicking any member name from the members list successfully retrieves and displays that member's full profile including their department, research group, role, and assigned tasks. We verified this through live testing and database inspection in phpMyAdmin.

←→↻🔍keelyfowler.rhody.dev/researchdb/Webapp/member_view.php?id=1☆📄⬇️K School⋮

110 TA MTH 215 CSC310 PHY203 CSF 202 211 TA TA Setup- - - Google...📏

📁 All Bookmarks

Alice Johnson – Research Group Database – Microsoft Internet Explorer

File Edit View Favorites Tools Help

⬅️➡️🔄🏠🔒

Back Forward Refresh Home Log Out

Address http://localhost/researchdb/member_view.php?id=1Go

 Alice Johnson

Logged In as: Alice Johnson
Role: Principal Investigator

🟢 Experiments

📁 Projects

👤 Members

📖 Literature

✅ Tasks

🏢 Departments

⬅️ Back to Members

✏️ Edit Member

🗑️ Delete

 Member Details

Name: Alice JohnsonEmail: alice.johnson@uni.edu

Role: Principal Investigator

🏢 Departments	👤 Research Groups	✅ Assigned Tasks				
Department	Group	Role	Joined	Title	Status	Progress
Biology	Genomics Lab	Principal Investigator	2020-01-10	Design sgRNA sequences	Completed	100%

✓ Done

research_group_db @ localhost

🌐 Local intranet

Part III: Documentation

A. Revision Explanations Summary

Revision	Why It Was Made	How It Improved the Project
Global search (search.php) with member search and status filters	Three reviewers noted search required CLI or SQL knowledge, which is not realistic for research users.	Users can now search across all five entity types from any page, and filter projects and tasks by status.
Performance indexes on status, date, and journal columns	Listed as a future improvement in Phase II final report. Search made these columns more critical to index.	Eliminates full table scans on frequently filtered columns and implements correct database design practice.
Fix project_access_view OR logic bug	Phase II reviewer flagged view logic should be verified. OR condition confirmed to make Project Lead branch unreachable.	View now correctly distinguishes standard members from Project Leads, matching the intended access control model.
Fix missing FROM keyword in member_view.php	Discovered during code audit; missing FROM keyword causes fatal SQL error on member profile pages.	Member profile pages load correctly and reliably; confirmed via live testing.

B. Updated Database Evidence

The following changes were made to research_group_db_final.sql:

- Six CREATE INDEX statements were added after the relevant table definitions for project, task, and literature
- The project_access_view CREATE VIEW statement was replaced with the corrected UNION-based definition
- The corresponding /*!50001 VIEW definition at the bottom of the SQL file was updated to match
- View definitions for member_contact_view and project_access_view were moved to after the last CREATE TABLE statement to fix a fatal import ordering error

Screenshots of the updated table structures showing the new indexes and the corrected view definition are in Part II above. These can be verified by importing research_group_db_final.sql and running SHOW INDEX FROM project, SHOW INDEX FROM task, and SHOW INDEX FROM literature in phpMyAdmin.

C. Updated Query/Functionality Evidence

The search feature is accessible at [search.php](#) after logging in. Submitting a search term runs five prepared statements across project, task, Person, experiment, and literature, with extended field coverage including year, role, objective, and description. Results are returned in grouped collapsible sections with result counts. Rows link directly to existing detail view pages. The status filter dropdowns on [projects.php](#) and [tasks.php](#) use parameterized GET queries and submit without JavaScript.

The search bar added to the quick-links nav bar on every page submits via a plain HTML GET form, meaning it functions without JavaScript and preserves the search term in the URL. Screenshots demonstrating the full search interface, collapsible result sections, status filters, and member profile fix are included in Part II above